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PRITHWISH SEN

Education

- 2022–present **PhD, Computer Science & Engineering**, *National Institute of Technology, Silchar*.
Design of Robust Chronic Wound Assessment Prognostic System using Deep Learning. **(Thesis Submitted)**
CPI: 9.2/10
- 2019–2021 : **MTech, Computer Science & Engineering**, *Indian Institute of Information Technology Guwahati*.
CPI : 9.15/10
- 2014–2018 : **B.E., Computer Science & Engineering**, *Jorhat Engineering College, Assam*.
Percentage : 73.9%

Publications

Journal Articles

- 2026 **Prithwish Sen**, and Pinki Roy. An energy-aware, load-balanced and secure iomt edge-cloud framework for real-time wound assessment, *Internet of Things*, volume 38, page 101965. Elsevier, 2026. **[Q1 SCIE Impact Factor: 7.1]**.
- 2025 **Prithwish Sen**, and Pinki Roy. Yolo meets ccvit-A lightweight end-to-end system for wound tissue analysis, *Biomedical Signal Processing and Control*, volume 106, page 107734. Elsevier, 2025. **[Q1 SCIE Impact Factor: 5.7]**.

Communicated Journals

- 2026 **Prithwish Sen**, Pinki Roy, Ananya Dutta Roy, Ratnam Deb Kanungo, Aparajeeta Paul. ChronicWoundDB: A Curated Clinical Dataset for Remote Chronic Wound Triage with Minimal Feature Benchmarking, In *Biomedical Signal Processing and Control* **[Q1 SCIE Impact Factor: 5.7]**.
- 2026 **Prithwish Sen**, Pinki Roy, Jiddu Krishnan O P and Arpita Choudhury. Depth-Aware Relative Surface Area Estimation of Chronic Wounds Using Monocular Depth and Segmentation, In *Journal of Medical Systems*. **[Q1 SCIE Impact Factor: 8.0]**.

In Conference Proceedings

- 2021 Prithwish Sen, Anindita Das, and Nilkanta Sahu. Rendering scenes for simulating adverse weather conditions. In *International Work-Conference on Artificial Neural Networks*, pages 347–358. Springer, 2021.
- 2021 Prithwish Sen, Anindita Das, and Nilkanta Sahu. Object detection in foggy weather conditions. In *International Conference on Intelligent Computing & Optimization*, pages 728–737. Springer, 2021.
- 2021 Prithwish Sen, Anindita Das, and Nilkanta Sahu. End-to-end scene text recognition system for devanagari and bengali text. In *International Conference on Intelligent Computing & Optimization*, pages 352–359. Springer, 2021.

- 2021 Anindita Das, Prithwish Sen, and Nilkanta Sahu. Adaptive style transfer using sirs. In *International Conference on Computer Analysis of Images and Patterns*, pages 352–361. Springer, 2021.

Patent Filed

- 2026 Ramanujam E, Prithwish Sen,. *TATDLO - Timer-based Automated Toilet Door-Latch Opener for the welfare of the Elderly patients (application under process)*

Research Projects

- June, 2022 – present ***Design of Robust Chronic Wound Assessment Prognostic System using Deep Learning. (PhD Project).***

Design of a deep learning based lightweight tissue analysis system and development of an IoT-enabled remote monitoring framework. Moreover, retrospective clinical dataset is curated for in-depth analysis and derived a minimal, high-impact feature set for wound prognosis, enabling efficient, interpretable, and resource-aware clinical decision support.

Advisor : **Dr. Pinki Roy**, Associate Professor, Department of Computer Science & Engineering, NIT Silchar ([Personal Web-page](#))

- May, 2021 – June, 2022 ***Design and Development of an Assistive Device for Persons with Motor Neuron Disease, TIDE, DST, Govt. of India (JRF).***

The task involves connecting brain and IoT device all together with the help of EEG device. The acquired signals are trained with different machine learning algorithms to infer the signals.

Advisor : **Dr. Kaustav Nag**, Assistant Professor, Department of Computer Science & Engineering, IIIT Guwahati ([Personal Web-page](#))

- June, 2020 – April, 2021 ***Object Detection In Adverse Weather Condition (Mtech Project).***

Proposed an approach for simulating realistic adverse weather conditions in order to synthesize adverse weather dataset via monocular depth-aware perlin noise integration for foggy weather. Additionally, an adverse weather Object detection framework have been developed using deep learning.

Advisor : **Dr. Nilkanta Sahu**, Assistant Professor, Department of Computer Science & Engineering, IIIT Guwahati ([Personal Web-page](#))

- Jan, 2020 – June, 2020 ***Multilingual Optical Character Recognition for scene text (Mtech Mini Project).***

This research project introduces a UNet-CNN framework for multilingual Indian scene text recognition. The U-Net specifically performs precise segmentation to remove the 'matra line', isolating character structures from the 'matra'. This refined output enables the CNN to accurately decode diverse scripts, significantly enhancing OCR performance in complex, real-world visual environments.

Advisor : **Dr. Nilkanta Sahu**, Assistant Professor, Department of Computer Science & Engineering, IIIT Guwahati ([Personal Web-page](#))

- Jan, 2020 – June, 2020 ***Instant Cool (Beverage Dispenser) (Mtech Mini Project).***

This project is about a smart vending machine that serves cold drinks automatically. It utilizes a Thermo-Electric cooling module to maintain the liquid at a refreshing temperature and an ESP8266 microcontroller to manage the dispensing process. The system is powered by IoT device, where a remote server sends a command via the MQTT protocol to trigger the pour. Once a user completes a transaction through a dedicated mobile app the user is allowed to pour the chilled liquid. By combining secure digital payments with real-time hardware control, the project offers a fully unmanned and efficient way to serve cold drinks.

Advisor : **Dr. Rakesh Matam**, Associate Professor, Computer Science & Engineering, IIIT Guwahati ([Personal Web-page](#))

July, 2017 – **A Lucy a Smart Home Automation System based on Raspberry Pi (Btech Project).**

May, 2018 This IoT-driven assistive device redefines smart home automation by integrating facial recognition with personalized conversation, delivering a responsive experience similar to Alexa, Siri, or Google Assistant. Beyond biometric identification, the system empowers users to control household appliances and conduct web searches using simple voice commands. It further enhances daily life by providing real-time weather updates and streaming music via YouTube on request. By combining natural language processing with remote hardware control, this project offers a seamless, hands-free solution that bridges the gap between digital intelligence and physical home management, making modern living more accessible through a familiar, intuitive AI persona.

Advisor : **Rajiv Chakraborty, Dr. Sauravjyoti Sarmah**, Assistant Professor, Computer Science & Engineering, Jorhat Engineering college ([Web-page](#))

Fellowships & Awards

2022 –2026 **MHRD Fellowship**, Government of India, as a PhD research scholar (SRF) in National Institute of Technology Silchar.

Academic Achievements & Recognitions

2021 1th rank in Master of Technology. IIIT Guwahati.

2018 10th rank in Bachelor of Engineering. Jorhat Engineering College.

Guest Faculty Experience

Oct– Nov, 2025 PG-DAI program at CDAC Guwahati (on-site).

Feb 2023 9 days training program for employees of SAIL, Bokaro (on-site).

Skills

Programming Languages Python, keras, Tensorflow, C, JAVA
Database SQL, MongoDB
Tools VMware, Matlab, Power BI, Tableau

Position of Responsibility

2020–2021 Former DPPC member at IIIT Guwahati.

2019–2020 coordintor of ROBOTICS CLUB IIIT Guwahati.

Referees

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